

NON-ABELIAN BOST-CONNES GROUPOIDS: TWO NO-GO THEOREMS, AND A CONSTRUCTION THAT REALIZES $\text{Prob}(G/N_\beta)$ EXACTLY WHEN THE FROBENIUS LIFTS MIX

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ABSTRACT. For a compact group G with Frobenius conjugacy classes $\text{Frob}_{\mathfrak{p}}$, $\mathfrak{p} \in S$, the Tannakian subgroup $N_\beta = \bigcap \{\ker \rho : \Xi_\beta(\rho, S) < \infty\}$ of Paper VI is the canonical non-abelian replacement of the obstruction group, and Paper VII showed that the obvious groupoid model, in which the Frobenius elements act through a homomorphism of the abelian ideal group, has KMS_β simplex $\text{Prob}(E_\beta \backslash G)$ with E_β abelian, which is $\text{Prob}(G/N_\beta)$ only when E_β is normal. We examine the two routes proposed in Paper VI for closing the gap and settle both.

Noncommutative coefficient algebras (Route B) cannot work, for a reason independent of the representation: a KMS_β state of $\mathcal{B} \rtimes J_S$ is tracial on $\mathcal{B}$, so the simplex is the simplex of scaling traces of $\mathcal{B}$, inner twists are invisible, and for $\mathcal{B} = C(Y_S) \otimes M_n(\mathbb{C})$ with any inner twist the equilibrium simplex is the abelian one.

Groupoids with isotropy N_β (the first reading of Route A) cannot work either: by Neshveyev's theorem the isotropy contributes traces on $C^*(N_\beta)$, i.e. *dual* data indexed by $\text{Irr}(N_\beta)$, and the extreme point given by the trivial representation is fixed by every symmetry, so the symmetry group never acts transitively on the extreme points, as it must for $\text{Prob}(G/N_\beta)$.

Principal groupoids with a non-abelian Frobenius cocycle (the second reading of Route A) do work, but not canonically. Ordering the primes by norm, the ordered products $P_n(x) = \text{Frob}_{\mathfrak{p}_1}^{x_1} \cdots \text{Frob}_{\mathfrak{p}_n}^{x_n}$ define a cocycle c on the tail relation with genuinely non-commuting increments, for any choice of lifts of the classes to elements. We construct the corresponding étale groupoid, prove that the KMS_β simplex of its corner algebra is $\text{Prob}(H \backslash G)$ with H the Mackey range of c , and prove a *non-abelian Kakutani theorem*: $\rho \circ c$ is a coboundary if and only if $\Xi_\beta(\rho, S) = \sum_{\mathfrak{p}} N_{\mathfrak{p}}^{-\beta} \|I - \rho(\text{Frob}_{\mathfrak{p}})\|_{\text{HS}}^2 < \infty$, the proof being a matrix martingale whose L^2 -growth is governed by $\det(I + t(I - U)^*(I - U)/(1 - t)^2)$. Consequently N_β is the normal closure of H , and the model realizes $\text{Prob}(G/N_\beta)$ precisely when H is normal. A mixing criterion on the products of the means $\mathbb{E}[\rho(\text{Frob}_{\mathfrak{p}})^{x_{\mathfrak{p}}}]$ forces $H = G$; for $G = S_3$ with the transposition class at every prime, the constant lift (12) gives three KMS_β states while the alternating lift (12), (13), (12), ... gives one. The simplex therefore depends on the lift, the lift-independent content is exactly N_β , and there is no canonical non-abelian Bost-Connes groupoid; but whenever N_β has no proper subgroup with normal closure N_β — for instance $N_\beta = A_3$, realized by 3-cycle classes on a divergent set and transposition classes on a convergent one — the simplex is $\text{Prob}(G/N_\beta)$ for every lift.

CONTENTS

1.	The question, and what is already known	2
2.	Route B: noncommutative coefficient algebras	2
3.	Route A, first reading: isotropy cannot carry N_β	4
4.	Route A, second reading: the ordered Frobenius cocycle	5
4.1.	The cocycle	5
4.2.	The groupoid and its algebra	5
4.3.	Classification by the Mackey range	6
4.4.	The non-abelian Kakutani theorem	6

Sources, code and data for the entire series: <https://github.com/Ruqing1963/localized-bost-connes>; this paper is in `papers/XXVII-nonabelian-groupoids/`.

4.5. When is the Mackey range normal? A mixing criterion	8
4.6. When the lift does not matter	9
5. Verdict	9
References	10

1. THE QUESTION, AND WHAT IS ALREADY KNOWN

Throughout, K is a number field, S an infinite set of finite primes with $\zeta_{K,S}(\beta) = \infty$, i.e. $\sum_{\mathfrak{p} \in S} N\mathfrak{p}^{-\beta} = \infty$, listed by non-decreasing norm as $\mathfrak{p}_1, \mathfrak{p}_2, \dots$, with $t_i = N\mathfrak{p}_i^{-\beta} \in (0, t_{\max}]$, $t_{\max} = 2^{-\beta} < 1$. We write $V = \mathbb{N}^S$ for the space of valuation vectors with the product measure $\pi_\beta = \bigotimes_i \mu_{t_i}$, $\mu_t(k) = (1-t)t^k$, forced on every KMS_β state by [6, Lem. 2.10]; $\mathcal{R}_S$ is the tail equivalence relation on V (finitely many coordinates changed), ergodic by the zero-one law [6, Lem. 3.1]; $\mathcal{F}_{\leq n} = \sigma(x_1, \dots, x_n)$ and $\mathcal{F}_{>n} = \sigma(x_{n+1}, x_{n+2}, \dots)$.

G is a compact second countable group; for each i a conjugacy class $C_i \subset G$ is given (the Frobenius class at $\mathfrak{p}_i$, or a Satake class), and a *lift* is a choice of elements $u_i \in C_i$. For a unitary representation ρ of G put

$$(1.1) \quad \Xi_\beta(\rho, S) = \sum_i t_i \|I - \rho(u_i)\|_{\text{HS}}^2 = 2 \sum_i t_i (\dim \rho - \Re \text{Tr } \rho(u_i)),$$

which depends only on the classes, and let $\mathcal{T}_\beta = \{\rho : \Xi_\beta(\rho, S) < \infty\}$ and $N_\beta = \bigcap_{\rho \in \mathcal{T}_\beta} \ker \rho$; $\mathcal{T}_\beta = \text{Rep}(G/N_\beta)$ is the Tannakian theorem of Paper VI of the series, and N_β depends only on the classes.

The series. The foundational paper is [6]; Paper I [7] treats the factor type. Papers VI (the Tannakian obstruction) and VII (the Galois-twisted groupoid) are in preparation; we recall from them only what is needed and prove everything used.

The abelian theory has three ingredients: a *space* on which the ideal semigroup acts, whose KMS_β measures are the equilibrium states; a *symmetry group* permuting them; and a *cocycle* whose coboundary question is the Kakutani dichotomy. The simplex is $\text{Prob}(G_S/\Xi_\beta^\perp)$ with G_S acting transitively on the extreme points [6, Thm. 3.6]. The target of a non-abelian theory is a system whose KMS_β simplex is $\text{Prob}(G/N_\beta)$ with G acting transitively on the extreme points by translation, stabilizer N_β . The model of Paper VII — I_S acting on $V \times G$ through the homomorphism $a \mapsto \prod u_i^{a_i}$, which requires commuting lifts — gives $\text{Prob}(E_\beta \backslash G)$ with E_β contained in the closed abelian group generated by the u_i , and N_β is the normal closure of E_β . Two ways around this were proposed in Paper VI (its Remark 6.1) and left open in Paper VII: a groupoid built directly, whose unit space carries the tail relation and a G -action and whose isotropy realizes N_β ; and a noncommutative coefficient algebra carrying ρ , with the commutative Ore semigroup J_S kept. We take the second first, because its failure is the simplest.

2. ROUTE B: NONCOMMUTATIVE COEFFICIENT ALGEBRAS

Let $\mathcal{B}$ be a unital C^* -algebra with an action α of $J_S = \mathbb{N}^{(S)}$ by injective endomorphisms with hereditary ranges, $\alpha_a(1)$ a projection, so that Laca's dilation applies [1]: $\mathcal{B} \rtimes_\alpha J_S$ is the full corner $\mathbf{1}_{\mathcal{B}}(\tilde{\mathcal{B}} \rtimes I_S)\mathbf{1}_{\mathcal{B}}$ of the crossed product of the direct limit $\tilde{\mathcal{B}} = \varinjlim (\mathcal{B}, \alpha)$ by the automorphic extension of α to the group I_S ; it is generated by $\mathcal{B}$ and isometries μ_a with $\mu_a b \mu_a^* = \alpha_a(b)$, and the dynamics is $\sigma_t(\mu_a) = N a^{it} \mu_a$, $\sigma_t|_{\mathcal{B}} = \text{id}$. The gauge action of the compact group $\hat{I}_S$ gives the faithful conditional expectation $E : \mathcal{B} \rtimes_\alpha J_S \rightarrow \mathcal{B}$, the average of the gauge automorphisms; so “a tractable faithful expectation onto the coefficient algebra” exists for every $\mathcal{B}$. It does not help.

Write $T(\mathcal{B})$ for the tracial state space of $\mathcal{B}$, a Choquet simplex, and

$$T_\beta(\mathcal{B}, \alpha) = \{ \tau \in T(\mathcal{B}) : \tau \circ \alpha_a = \mathbf{N}a^{-\beta} \tau \quad (a \in J_S) \},$$

the *scaling traces*, a closed face of $T(\mathcal{B})$.

Theorem 2.1 (Route B is abelian). (1) Every KMS_β state φ of $\mathcal{B} \rtimes_\alpha J_S$ restricts to a scaling trace $\tau = \varphi|_{\mathcal{B}} \in T_\beta(\mathcal{B}, \alpha)$.

(2) For every $\tau \in T_\beta(\mathcal{B}, \alpha)$, $\tau \circ E$ is a KMS_β state with restriction τ ; so $\tau \mapsto \tau \circ E$ is an affine injection $T_\beta(\mathcal{B}, \alpha) \rightarrow \text{KMS}_\beta(\mathcal{B} \rtimes_\alpha J_S)$.

(3) If the numbers $\log \mathbf{N}p$, $p \in S$, are linearly independent over $\mathbb{Q}$ — for instance $K = \mathbb{Q}$ — then every KMS_β state is of the form $\tau \circ E$, and $\text{KMS}_\beta(\mathcal{B} \rtimes_\alpha J_S) \cong T_\beta(\mathcal{B}, \alpha)$ affinely.

Consequently: (a) if $\alpha'_a = \text{Ad}(w_a) \circ \alpha_a$ with unitaries $w_a \in \mathcal{B}$ is another action, then $T_\beta(\mathcal{B}, \alpha') = T_\beta(\mathcal{B}, \alpha)$; inner twists are invisible. (b) For $\mathcal{B} = C(Y_S) \otimes M_n(\mathbb{C})$ and any action of the form $\alpha_a \otimes \text{Ad}(v_a)$, $v_a \in U(n)$, the scaling traces are $\nu \otimes \frac{1}{n} \text{Tr}$ with ν a KMS_β measure of the classical system, so $T_\beta = \text{Prob}(G_S / \Xi_\beta^\perp)$ [6, Thm. 3.6]: the abelian answer. (c) In every case the equilibrium theory of $\mathcal{B} \rtimes_\alpha J_S$ is that of the affine action of J_S on the Choquet simplex $T(\mathcal{B})$; the noncommutativity of $\mathcal{B}$ is quotiented out by traciality before any symmetry can act on it.

Proof. (1) $\mathcal{B}$ lies in the fixed-point algebra of σ , and for x, y there the KMS condition gives $\varphi(xy) = \varphi(y\sigma_{i\beta}(x)) = \varphi(yx)$; so τ is tracial. For $b \in \mathcal{B}$, the KMS condition with the entire element μ_a , $\sigma_{i\beta}(\mu_a) = \mathbf{N}a^{-\beta} \mu_a$, gives $\tau(\alpha_a(b)) = \varphi(\mu_a b \mu_a^*) = \mathbf{N}a^{-\beta} \varphi(b \mu_a^* \mu_a) = \mathbf{N}a^{-\beta} \tau(b)$.

(2) Extend τ to the semifinite trace $\tilde{\tau}$ on $\tilde{\mathcal{B}}$ determined by $\tilde{\tau}(\alpha_a^{-1}(b)) = \mathbf{N}a^\beta \tau(b)$, consistent by the scaling relation, and let γ be the automorphic action of I_S on $\tilde{\mathcal{B}}$, so that $\tilde{\tau} \circ \gamma_a = \mathbf{N}a^{-\beta} \tilde{\tau}$ for $a \in I_S$. On the dense span of elements $x = a_1 u_a$, $y = a_2 u_b$ of $\tilde{\mathcal{B}} \rtimes I_S$ ($a_i \in \tilde{\mathcal{B}}$ in the domain of $\tilde{\tau}$, u_a the implementing unitaries),

$$(\tilde{\tau} \circ E)(xy) = \delta_{a+b,0} \tilde{\tau}(a_1 \gamma_a(a_2)), \quad (\tilde{\tau} \circ E)(y \sigma_{i\beta}(x)) = \delta_{a+b,0} \mathbf{N}a^{-\beta} \tilde{\tau}(a_2 \gamma_{-a}(a_1)),$$

and $\mathbf{N}a^{-\beta} \tilde{\tau}(a_2 \gamma_{-a}(a_1)) = \mathbf{N}a^{-\beta} \mathbf{N}a^\beta \tilde{\tau}(\gamma_a(a_2) a_1) = \tilde{\tau}(a_1 \gamma_a(a_2))$, using $\tilde{\tau} \circ \gamma_a = \mathbf{N}a^{-\beta} \tilde{\tau}$ read as $\tilde{\tau} = \mathbf{N}a^\beta \tilde{\tau} \circ \gamma_a$ and then traciality. So $\tilde{\tau} \circ E$ satisfies the KMS condition on the corner, where it is the state $\tau \circ E$. Injectivity of $\tau \mapsto \tau \circ E$ is clear.

(3) A KMS_β state is σ -invariant. Under the hypothesis, the one-parameter group $t \mapsto (\mathbf{N}p^{it})_p$ is dense in $\widehat{I_S} = \prod_p \mathbb{T}$ by Kronecker's theorem applied to finite subsets of S , so a σ -invariant state is invariant under the whole gauge action and hence factors through E : $\varphi = \varphi \circ E = \tau \circ E$ with $\tau = \varphi|_{\mathcal{B}}$, which is a scaling trace by (1).

(a) A trace is invariant under inner automorphisms, so $\tau \circ \alpha'_a = \tau \circ \alpha_a$. (b) A tracial state of $C(Y_S) \otimes M_n$ is $\nu \otimes \frac{1}{n} \text{Tr}$, and $\text{Tr}(v_a m v_a^*) = \text{Tr } m$ reduces the scaling condition to $\nu \circ \alpha_a = \mathbf{N}a^{-\beta} \nu$. (c) is (1)–(3). $\square$

Remark 2.2. Without the independence hypothesis in (3), σ -invariance only forces φ to factor through the expectation onto the fixed-point algebra of σ , which also contains the elements $\mu_a b \mu_b^*$ with $\mathbf{N}a = \mathbf{N}b$, $a \neq b$; for the classical system these carry nothing because the groupoid is principal on a conull set [6, Prop. 2.8], and for general $\mathcal{B}$ there may be additional KMS_β states, all of them tracial on the fixed-point algebra. None of this reintroduces the representation: the matrix Kakutani dichotomy of Paper VI, valid for commuting lifts, would enter Route B through the twist $\text{Ad } \rho(u_p)$, and (a) says the twist is never seen; for non-commuting lifts $\alpha_p \otimes \text{Ad } \rho(u_p)^{-1}$ do not even commute, so there is no action to twist. The modular group is not “broken” by a noncommutative $\mathcal{B}$: it is the same gauge dynamics, and it forces the equilibrium state to forget everything about $\mathcal{B}$ except its traces. Route B is closed.

3. ROUTE A, FIRST READING: ISOTROPY CANNOT CARRY N_β

Let $\mathcal{G}$ be a second countable locally compact Hausdorff étale groupoid with unit space Ω , and $c : \mathcal{G} \rightarrow \mathbb{R}$ a continuous cocycle defining $\sigma_t(f)(\gamma) = e^{itc(\gamma)}f(\gamma)$ on $C^*(\mathcal{G})$ (the Haar system is counting measure on the fibres, as for every étale groupoid). Neshveyev's theorem [3, Thm. 1.3] describes the KMS_β states: they are the functionals

$$(3.1) \quad \varphi(f) = \int_{\Omega} \sum_{\gamma \in \mathcal{G}_x^x} f(\gamma) \phi_x(u_\gamma) d\mu(x),$$

where μ is a quasi-invariant probability measure with Radon–Nikodym cocycle $e^{-\beta c}$ and $x \mapsto \phi_x$ is a μ -measurable field of tracial states on the group C^* -algebras $C^*(\mathcal{G}_x^x)$ of the isotropy groups, vanishing on u_γ unless $c(\gamma) = 0$, and conjugation-equivariant: $\phi_{r(\gamma)}(u_{\gamma h \gamma^{-1}}) = \phi_{s(\gamma)}(u_h)$ for μ -a.e. γ . The extreme points are the pairs with μ ergodic and ϕ extremal among equivariant fields.

Theorem 3.1 (No-go for isotropy). *Let $(\mathcal{G}, c)$ be as above, and let a group G act on $\mathcal{G}$ by automorphisms preserving c , hence on $\text{KMS}_\beta(C^*(\mathcal{G}))$ by affine bijections. For an extreme KMS_β state with quasi-invariant measure μ let $\mathcal{E}_\mu$ be the set of extreme KMS_β states with measure μ .*

- (1) *The state $\varphi_\mu(f) = \int_{\Omega} f|_{\Omega} d\mu$ — the field $\phi_x = \text{trivial character}$ — belongs to $\mathcal{E}_\mu$, and it is fixed by the stabilizer G_μ of μ in G .*
- (2) *If the isotropy groups $\mathcal{G}_x^x$ are μ -a.e. isomorphic to a fixed compact group $N \subseteq \ker c$, with the isotropy bundle constant and the conjugation action of $\mathcal{G}$ on it trivial, then $\mathcal{E}_\mu$ is in bijection with $\text{Irr}(N)$, via $\rho \mapsto (\mu, \phi_x \equiv \chi_\rho / \dim \rho)$.*
- (3) *Consequently, if G acts transitively on the extreme KMS_β states, then $\mathcal{E}_\mu = \{\varphi_\mu\}$ for every μ , and in the situation of (2) $N = \{1\}$: the isotropy is μ -a.e. trivial.*

In particular no groupoid whose KMS_β multiplicity is produced by isotropy has a simplex of the form $\text{Prob}(G/N)$ with G acting transitively on the extreme points and $N \neq 1$.

Proof. (1) The trivial character is a tracial state of every $C^*(\mathcal{G}_x^x)$, vanishes on u_γ for $c(\gamma) \neq 0$ as required, and the constant field is conjugation-equivariant; so φ_μ is a KMS_β state by (3.1), and it is extreme because μ is ergodic and the trivial character is an extreme tracial state. An automorphism of $(\mathcal{G}, c)$ transforms a pair (μ, ϕ) into $(\mu \circ g^{-1}, \phi \circ g^{-1})$; if it fixes μ it fixes the pair with the trivial field.

(2) Tracial states of $C^*(N)$ are the convex combinations of normalized characters $\chi_\rho / \dim \rho$, $\rho \in \text{Irr}(N)$, the extreme ones being the normalized characters. With constant bundle and trivial conjugation action, equivariance says the field is $\mathcal{G}$ -invariant, hence a.e. constant by ergodicity of μ ; so the equivariant fields are the constant ones, the extreme ones the constant normalized characters.

(3) The map $\varphi \mapsto \mu$ from extreme KMS_β states to ergodic quasi-invariant measures is G -equivariant, so if G is transitive on the extreme states then G_μ is transitive on the fibre $\mathcal{E}_\mu$; by (1) it fixes $\varphi_\mu \in \mathcal{E}_\mu$, so $\mathcal{E}_\mu = \{\varphi_\mu\}$, and by (2) $|\text{Irr}(N)| = 1$, i.e. $N = 1$. For the last sentence, the translation action of G on $\text{Prob}(G/N)$ is transitive on the extreme points. $\square$

Remark 3.2. The theorem says that isotropy is the wrong place to put N_β : it produces the *dual* object $\text{Irr}(N_\beta)$, with the trivial representation as a distinguished, symmetry-fixed equilibrium state, whereas symmetry breaking needs a homogeneous space with no distinguished point. “Isotropy = N_β ” would give a simplex with a canonical symmetric state and $|\text{Irr}(N_\beta)|$ extreme points over each ergodic measure, which is $\text{Prob}(G/N_\beta)$ neither as a G -space nor, in general, as a set. The classical system itself illustrates the point: its groupoid is principal on a conull set [6, Prop. 2.8], and all of its multiplicity comes from the ergodic decomposition of the unit space. That is where a non-abelian construction must put it.

4. ROUTE A, SECOND READING: THE ORDERED FROBENIUS COCYCLE

4.1. **The cocycle.** Fix lifts $u_i \in C_i$. For $x \in V$ and $n \geq 0$ put

$$(4.1) \quad P_n(x) = u_1^{x_1} u_2^{x_2} \cdots u_n^{x_n} \in G,$$

an ordered product, and for $(y, x) \in \mathcal{R}_S$ let n be any index beyond the last coordinate in which x and y differ and

$$(4.2) \quad c(y, x) = P_n(y) P_n(x)^{-1}.$$

This is independent of n : $P_{n'}(x) = P_n(x) u_{n+1}^{x_{n+1}} \cdots u_{n'}^{x_{n'}}$ and the extra factors agree for x and y and cancel on the right. It is a cocycle, $c(z, y)c(y, x) = c(z, x)$, being of the form $P(y)P(x)^{-1}$ on each $\mathcal{R}_{F_n}$. Its increments are

$$(4.3) \quad c(x + e_j, x) = P_{j-1}(x) u_j P_{j-1}(x)^{-1},$$

the j -th lift conjugated by the ordered product of the earlier ones; for commuting lifts this is u_j and c is the constant-increment cocycle of Paper VII, but in general the increments at different coordinates do not commute, and (4.2) is the way to have a cocycle nonetheless: the obstruction of Paper VI, that a cocycle with *constant* non-commuting increments cannot exist on a relation generated by commuting shifts, is circumvented by letting the increment at $\mathfrak{p}_j$ depend on the coordinates before it.

4.2. **The groupoid and its algebra.** To have a topological model we compactify each coordinate together with its lift. Let

$$X_i = \overline{\{(k, u_i^k) : k \in \mathbb{N}\}} \subseteq \overline{\mathbb{N}} \times G, \quad X = \prod_{i \geq 1} X_i,$$

compact, second countable, with $\mathbb{N} \subset X_i$ open and dense (as $\{k\} \times \{u_i^k\}$) and boundary $\partial X_i = \{\infty\} \times \overline{\{u_i^k\}}$. Write $g_i : X_i \rightarrow G$ for the second projection, continuous, with $g_i(k) = u_i^k$. The shift $k \mapsto k+1$ extends to a homeomorphism s_i of X_i onto $X_i \setminus \{0\}$, $(\infty, g) \mapsto (\infty, u_i g)$, and J_S acts on X by $x \mapsto x + a$ coordinatewise, injectively, with clopen images. Put

$$P_n(x) = g_1(x_1) \cdots g_n(x_n), \quad c(x + e_j, x) = P_{j-1}(x) u_j P_{j-1}(x)^{-1},$$

continuous on X and extending (4.1)–(4.3). Let $\tilde{X} = \varinjlim (X, +a)$ be the dilation, a locally compact space on which $I_S = \mathbb{Z}^{(S)}$ acts by homeomorphisms, and extend c to $\tilde{X}$ by the same formulas. Define the action of I_S on $\Omega = \tilde{X} \times G$ by

$$(4.4) \quad a \cdot (x, g) = (x + a, c(x + a, x) g),$$

which is an action because c is a cocycle, and by homeomorphisms because c is continuous. Let

$$\mathcal{G}_c = I_S \ltimes \Omega, \quad \mathcal{A}_c = \mathbf{1}_{X \times G} C^*(\mathcal{G}_c) \mathbf{1}_{X \times G} = C(X \times G) \rtimes J_S,$$

the transformation groupoid with counting measures as Haar system [4], and its unital full corner; the dynamics is $\sigma_t(\mu_a) = \mathbf{N} a^{it} \mu_a$, given by the continuous cocycle $(a, \omega) \mapsto \log \mathbf{N} a$ on $\mathcal{G}_c$. The group G acts on Ω by right translation $R_h(x, g) = (x, gh)$, commuting with (4.4), hence on $\mathcal{A}_c$ by σ -commuting automorphisms.

Lemma 4.1. $\mathcal{G}_c$ is second countable, étale, Hausdorff and amenable, principal on the conull set $V \times G$ (points with all coordinates finite), and $\mathcal{A}_c$ is nuclear. The KMS_β states of $\mathcal{A}_c$ are in affine bijection with the probability measures ν on $X \times G$ with $\nu(a \cdot E) = \mathbf{N} a^{-\beta} \nu(E)$ for $a \in J_S$; every such ν is concentrated on $V \times G$, has V -marginal π_β , and disintegrates as $\nu = \pi_\beta \otimes \rho_x$ with a measurable family $\rho_x \in \text{Prob}(G)$ satisfying

$$(4.5) \quad \rho_{x+e_j} = (c(x + e_j, x))_* \rho_x \quad \text{for } \pi_\beta\text{-a.e. } x, \text{ every } j.$$

Proof. I_S is abelian, so $\mathcal{G}_c$ is amenable and $C^* = C_r^*$; étaleness and Hausdorffness are those of a transformation groupoid by homeomorphisms. If $a \cdot (x, g) = (x, g)$ then $x + a = x$, which forces $a = 0$ when all x_i are finite; so the isotropy is trivial on $V \times G$. For the KMS states, the argument of [6, Prop. 2.8] applies verbatim (or one may invoke (3.1)): a KMS_β state is $\nu \circ E$ for a scaling measure ν on the unit space of the corner. If $E \subseteq \{x_i = \infty\}$ then $e_i \cdot E = E$ and the scaling gives $\nu(E) = t_i \nu(E)$, so $\nu(E) = 0$: ν is concentrated on $V \times G$. The V -marginal satisfies the scaling relations of the valuation coordinates alone, so it is π_β by [6, Lem. 2.10]. Disintegrating and comparing the two sides of the scaling identity on sets $B \times C$, using $\pi_\beta(B + e_j) = t_j \pi_\beta(B)$, gives (4.5). $\square$

4.3. Classification by the Mackey range. Recall Zimmer's theory of cocycles into compact groups [5, Ch. 4 and 9], in the setting of countable equivalence relations [2]: for an ergodic countable equivalence relation $(V, \pi, \mathcal{R})$ and a measurable cocycle $c : \mathcal{R} \rightarrow G$ into a compact group, there is a closed subgroup $H \leq G$, unique up to conjugacy, and a measurable $f : V \rightarrow G$ such that $c'(y, x) = f(y)^{-1} c(y, x) f(x)$ takes values in H and the skew product relation $\mathcal{R} \ltimes_{c'} H$ on $V \times H$ is ergodic; H is the *Mackey range* (algebraic hull) of c , and c is cohomologous to a cocycle into a closed subgroup H' iff H' contains a conjugate of H .

Theorem 4.2 (The simplex of the ordered model). *Let H be the Mackey range of c on $(V, \pi_\beta, \mathcal{R}_S)$. Then*

$$\text{KMS}_\beta(\mathcal{A}_c) \cong_{\text{affine}} \text{Prob}(H \backslash G),$$

the extreme points being the families $\rho_x = \text{Haar measure on } f(x)Hk$, $k \in G$, and the right translation action of G is transitive on the extreme points with stabilizer of the point Hk equal to $k^{-1}Hk$.

Proof. By Lemma 4.1 we classify the equivariant families (4.5), i.e. the probability measures $\nu = \pi_\beta \otimes \rho_x$ on $V \times G$ invariant in measure class and with the prescribed Radon–Nikodym derivative under the skew product relation $\tilde{\mathcal{R}} = \mathcal{R}_S \ltimes_c G$, $(x, g) \sim (y, c(y, x)g)$; the extreme ones are the ergodic ones. The map $\Psi(x, g) = (x, f(x)^{-1}g)$ carries $\tilde{\mathcal{R}}$ to $\mathcal{R}_S \ltimes_{c'} G$ and preserves $\pi_\beta \otimes m_G$. The latter relation preserves each set $V \times Hk$, on which it is isomorphic to $\mathcal{R}_S \ltimes_{c'} H$ via $h \mapsto hk$, ergodic by the definition of the Mackey range. Hence the ergodic components of $\tilde{\mathcal{R}}$ are the sets $\Psi^{-1}(V \times Hk) = \{(x, g) : f(x)^{-1}g \in Hk\}$, $Hk \in H \backslash G$. Within a component, Haar on the fibres is the *only* equivariant family: if $\rho'_x \in \text{Prob}(H)$ satisfies $\rho'_y = c'(y, x)_* \rho'_x$, then for a nontrivial $\sigma \in \text{Irr}(H)$ the Fourier coefficient $B(x) = (\int_H \sigma(h)^* d\rho'_x(h))^*$ satisfies $B(y) = \sigma(c'(y, x))B(x)$, so $\Phi(x, h) = \text{Tr}(\sigma(h)^* B(x))$ is $\mathcal{R}_S \ltimes_{c'} H$ -invariant — $\Phi(y, c'h) = \text{Tr}(\sigma(h)^* \sigma(c')^* \sigma(c') B(x)) = \Phi(x, h)$ — hence a.e. constant by ergodicity; since $\int_H \sigma(h) dh = 0$ the constant is 0, and Fourier inversion in h gives $B = 0$. All nontrivial coefficients of ρ'_x vanish, so $\rho'_x = m_H$ a.e. Thus the ergodic equivariant families are the Haar measures on the fibres $f(x)Hk$, and every equivariant family is a probability mixture of these, uniquely, since distinct ergodic components carry mutually singular measures. Right translation by h maps the component of Hk to that of Hkh . $\square$

For commuting lifts f can be taken trivial and $H = E_\beta$, recovering Paper VII. The content of the model is therefore the conjugacy class of H . Two questions remain: how H is related to N_β , and how H depends on the lifts.

4.4. The non-abelian Kakutani theorem. For a unitary representation $\rho : G \rightarrow U(d)$ write $U_i = \rho(u_i)$ and

$$(4.6) \quad m_i = \mathbb{E}_{\mu_{t_i}}[U_i^{x_i}] = \sum_{k \geq 0} (1 - t_i) t_i^k U_i^k = (1 - t_i)(I - t_i U_i)^{-1},$$

invertible since $\|t_i U_i\| < 1$, with

$$(4.7) \quad m_i^{-*} m_i^{-1} = \frac{(I - t_i U_i^*)(I - t_i U_i)}{(1 - t_i)^2} = I + \frac{t_i}{(1 - t_i)^2} (I - U_i)^*(I - U_i) \geq I.$$

If $e^{i\phi_{i,1}}, \dots, e^{i\phi_{i,d}}$ are the eigenvalues of U_i , then

$$(4.8) \quad \det(m_i^{-*} m_i^{-1}) = \prod_{r=1}^d \left(1 + \frac{t_i |1 - e^{i\phi_{i,r}}|^2}{(1 - t_i)^2}\right), \quad \sum_r |1 - e^{i\phi_{i,r}}|^2 = \|I - U_i\|_{\text{HS}}^2.$$

Put $M_{(n,m]} = m_{n+1} m_{n+2} \cdots m_m$ and $M_n = M_{(0,n]}$. A *coboundary* means a measurable $F : V \rightarrow U(d)$ with $\rho(c(y, x)) = F(y)F(x)^{-1}$ for π_β -a.e. $(y, x) \in \mathcal{R}_S$.

Theorem 4.3 (Non-abelian Kakutani). *For every lift and every unitary representation ρ ,*

$$\rho \circ c \text{ is a coboundary} \iff \Xi_\beta(\rho, S) < \infty.$$

Proof. Two structural facts are used throughout. First, for $F : V \rightarrow M_d(\mathbb{C})$ measurable with $F(y) = \rho(c(y, x))F(x)$ on $\mathcal{R}_S$, the function $K_n = P_n^{-1}F$ (with $P_n = \rho(P_n(x))$, we suppress ρ) is invariant under changes of the first n coordinates: for $y \sim x$ differing only there, $K_n(y) = P_n(y)^{-1}\rho(c(y, x))F(x) = P_n(y)^{-1}P_n(y)P_n(x)^{-1}F(x) = K_n(x)$; so K_n is $\mathcal{F}_{>n}$ -measurable [6, Lem. 3.1] and independent of P_n , which is $\mathcal{F}_{\leq n}$ -measurable. Second, $\mathbb{E}[P_n] = M_n$ and $\mathbb{E}[P_m | \mathcal{F}_{\leq n}] = P_n M_{(n,m]}$, by independence of the coordinates and the ordering of the product.

($\Leftarrow$) Let $\Xi_\beta(\rho, S) < \infty$ and $Z_n = P_n M_n^{-1}$. Then $\mathbb{E}[Z_{n+1} | \mathcal{F}_{\leq n}] = P_n m_{n+1} M_{n+1}^{-1} = Z_n$: a matrix-valued martingale. Its L^2 -norm is deterministic, $\mathbb{E}\|Z_n\|_{\text{HS}}^2 = \text{Tr}(M_n^{-1} M_n^{-*}) = \|M_n^{-1}\|_{\text{HS}}^2$, and by (4.7), iterating $m_i^{-*}(\cdot) m_i^{-1} \leq \lambda_i(\cdot)$ with $\lambda_i = 1 + t_i \|I - U_i\|_{\text{op}}^2 / (1 - t_i)^2$ from the inside out,

$$\|M_n^{-1}\|_{\text{HS}}^2 \leq d \prod_{i \leq n} \lambda_i \leq d \exp\left(\frac{1}{(1 - t_{\max})^2} \sum_i t_i \|I - U_i\|_{\text{HS}}^2\right) < \infty,$$

using $t_i \leq t_{\max} < 1$. So $Z_n \rightarrow Z_\infty =: F$ a.s. and in L^2 . Moreover $|\det Z_n| = |\det M_n^{-1}| = \prod_{i \leq n} \det(m_i^{-*} m_i^{-1})^{1/2}$ increases to a finite limit ≥ 1 by (4.8) and $\Xi_\beta < \infty$, so $\det F \neq 0$ a.s. and F is invertible. For $y \sim x$ and n large, $Z_n(y)Z_n(x)^{-1} = P_n(y)P_n(x)^{-1} = \rho(c(y, x))$, and letting $n \rightarrow \infty$ gives $F(y) = \rho(c(y, x))F(x)$. Then F^*F is $\mathcal{R}_S$ -invariant, $F(y)^*F(y) = F(x)^*\rho(c)^*\rho(c)F(x) = F(x)^*F(x)$, hence a.e. equal to a constant positive matrix Q^2 , and $W = FQ^{-1}$ is unitary with $\rho(c(y, x)) = W(y)W(x)^{-1}$.

($\Rightarrow$) Let F be a unitary coboundary and $C_n = \mathbb{E}[K_n]$. Then $\mathbb{E}[F | \mathcal{F}_{\leq n}] = P_n C_n$, and comparing $\mathbb{E}[F | \mathcal{F}_{\leq n}] = \mathbb{E}[\mathbb{E}[F | \mathcal{F}_{\leq m}] | \mathcal{F}_{\leq n}] = P_n M_{(n,m]} C_m$ gives

$$(4.9) \quad C_n = M_{(n,m]} C_m \quad (m > n).$$

Suppose $\Xi_\beta(\rho, S) = \infty$. Then for each n , $|\det M_{(n,m]}^{-1}| \rightarrow \infty$ as $m \rightarrow \infty$ by (4.8), while $\|C_m\|_{\text{HS}} \leq \mathbb{E}\|K_m\|_{\text{HS}} = \sqrt{d}$; so $|\det C_m| = |\det M_{(n,m]}^{-1}| |\det C_n|$ bounded forces $\det C_n = 0$ for every n . But $P_n C_n = \mathbb{E}[F | \mathcal{F}_{\leq n}] \rightarrow F$ in L^2 by martingale convergence, F being measurable with respect to $\bigvee_n \mathcal{F}_{\leq n}$, so along a subsequence $\det(P_{n_k} C_{n_k}) \rightarrow \det F$ a.e., and $|\det F| = 1$; this contradicts $\det C_{n_k} = 0$. $\square$

Corollary 4.4 (N_β is the normal closure of the Mackey range). *For every lift, with H the Mackey range of c ,*

$$\mathcal{T}_\beta = \{\rho : \rho|_H = 1\}, \quad N_\beta = \langle\langle H \rangle\rangle,$$

the smallest closed normal subgroup containing H . Consequently $\text{KMS}_\beta(\mathcal{A}_c) = \text{Prob}(G/N_\beta)$, with G acting on the extreme points as on G/N_β , if and only if H is normal in G ; in general there is a G -equivariant surjection $H \setminus G \rightarrow G/N_\beta$ of extreme boundaries, a bijection iff $H \trianglelefteq G$.

Proof. $\rho \circ c$ is a coboundary iff the Mackey range of $\rho \circ c$ is trivial; and the Mackey range of $\rho \circ c$ is $\rho(H)$, since $\rho \circ c'$ takes values in $\rho(H)$ and the skew product by $\rho \circ c'$ on $V \times \rho(H)$ is a factor of the ergodic $\mathcal{R}_S \ltimes_c H$. So by Theorem 4.3, $\Xi_\beta(\rho, S) < \infty$ iff $\rho|_H = 1$. The intersection of the kernels of the representations trivial on H is the closed normal subgroup generated by H , by Peter–Weyl applied to $G/\langle\langle H \rangle\rangle$. The rest is Theorem 4.2. $\square$

4.5. When is the Mackey range normal? A mixing criterion.

Theorem 4.5 (Mixing forces $H = G$). *Suppose that for every nontrivial irreducible ρ and every n , $\|M_{(n,m)}\|_{\text{op}} \rightarrow 0$ as $m \rightarrow \infty$. Then the skew product $\mathcal{R}_S \ltimes_c G$ is ergodic, $H = G$, and $\mathcal{A}_c$ has a unique KMS_β state.*

Proof. Let $\Phi \in L^\infty(V \times G)$ be $\tilde{\mathcal{R}}$ -invariant and $A_\rho(x) = \int_G \Phi(x, h) \rho(h)^* dh$ its Fourier coefficients. Invariance gives $A_\rho(y) = A_\rho(x) \rho(c(y, x))^*$, so $B = A_\rho^*$ satisfies $B(y) = \rho(c(y, x)) B(x)$ and is bounded. As in the proof of Theorem 4.3, $\mathbb{E}[B | \mathcal{F}_{\leq n}] = P_n C_n$ with $C_n = \mathbb{E}[P_n^{-1} B]$ satisfying (4.9) and $\|C_m\|_{\text{HS}} \leq \sqrt{d} \|B\|_\infty$. By hypothesis $\|C_n\| \leq \|M_{(n,m)}\|_{\text{op}} \sup_m \|C_m\| \rightarrow 0$, so $C_n = 0$ for all n , hence $B = 0$ by martingale convergence: $A_\rho = 0$ for every nontrivial irreducible ρ . Thus $\Phi(x, g)$ does not depend on g , and then it is $\mathcal{R}_S$ -invariant, hence constant. Ergodicity of the skew product means $H = G$, and Theorem 4.2 gives the rest. $\square$

The hypothesis fails exactly when some nontrivial ρ has a direction that the contractions m_i asymptotically leave alone — an H -fixed vector. The criterion is easy to check in examples, and it is what separates the lifts.

Theorem 4.6 (The simplex depends on the lift). *Let $G = S_3$, and let the class C_i be the class of transpositions for every i .*

- (1) *For the constant lift $u_i = (12)$, $H = \{e, (12)\}$ and $\mathcal{A}_c$ has exactly three extremal KMS_β states, permuted transitively by S_3 with stabilizers of order 2.*
- (2) *For the alternating lift $u_{2k-1} = (12)$, $u_{2k} = (13)$, $H = S_3$ and $\mathcal{A}_c$ has a unique KMS_β state.*

In both cases $N_\beta = S_3$ and $\text{Prob}(G/N_\beta)$ is a point: the Tannakian prediction is right for the second lift and wrong for the first.

Proof. (1) The lifts commute, c has constant increments (12) , and H is the Mackey range of the constant cocycle into the abelian group $A = \{e, (12)\}$, namely the annihilator of the characters χ of A with $\sum_i t_i |1 - \chi((12))|^2 < \infty$; the nontrivial character has $|1 - \chi((12))|^2 = 4$ and $\sum t_i = \infty$, so only the trivial character qualifies and $H = A$. Theorem 4.2 gives $\text{Prob}(A \setminus S_3)$, three points.

(2) We verify the hypothesis of Theorem 4.5. For the sign character, $m_i = (1 - t_i)/(1 + t_i)$ and $\prod_{i>n} m_i = 0$ since $\sum t_i = \infty$. For the two-dimensional irreducible ρ , $\rho(u_i)$ is a reflection with fixed line ℓ_i , ℓ_{2k-1} and ℓ_{2k} making an angle of 60° ; by (4.6), $m_i = Q_i + \lambda_i Q_i^\perp$ with Q_i the projection onto ℓ_i and $\lambda_i = (1 - t_i)/(1 + t_i)$. For a unit vector v either $\|Q_i^\perp v\| \geq \frac{1}{2}$ or $\|Q_{i+1}^\perp v\| \geq \frac{1}{2}$, the two lines being 60° apart. In the first case $\|m_i v\|^2 \leq 1 - \frac{1}{4}(1 - \lambda_i^2)$. In the second, $\|m_i v - v\| \leq 1 - \lambda_i \leq 2t_i \leq \frac{1}{4}$ once $t_i \leq \frac{1}{8}$, so $\|Q_{i+1}^\perp m_i v\| \geq \frac{1}{4}$ and $\|m_{i+1} m_i v\|^2 \leq 1 - \frac{1}{16}(1 - \lambda_{i+1}^2)$. As $1 - \lambda_i^2 \geq t_i$, in both cases $\|m_{i+1} m_i\|_{\text{op}} \leq 1 - \frac{1}{32} \min(t_i, t_{i+1})$, so

$$\|M_{(n,m)}\|_{\text{op}} \leq \prod_{k: n < 2k-1 < 2k \leq m} \left(1 - \frac{1}{32} \min(t_{2k-1}, t_{2k})\right) \leq \exp\left(-\frac{1}{32} \sum_k \min(t_{2k-1}, t_{2k})\right) \rightarrow 0,$$

because the t_i are non-increasing, so $\min(t_{2k-1}, t_{2k}) = t_{2k} \geq t_{2k+1}$ and $\sum_k t_{2k} \geq \frac{1}{2} \sum_{i \geq 2} t_i = \infty$. Theorem 4.5 applies.

For N_β : $\Xi_\beta(\text{sgn}, S) = 4 \sum t_i = \infty$ and $\Xi_\beta(\rho, S) = \sum t_i \|I - \rho(u_i)\|_{\text{HS}}^2 = 4 \sum t_i = \infty$, so $\mathcal{T}_\beta = \{\mathbf{1}\}$ and $N_\beta = S_3$ for either lift, as it must, N_β depending only on the classes. $\square$

Corollary 4.7 (No canonical non-abelian model). *The KMS_β simplex of $\mathcal{A}_c$ is not a function of the Frobenius classes $(C_i)_i$: it depends on the lift, through the conjugacy class of the Mackey range H . The lift-independent invariants of the construction are exactly those of $N_\beta = \langle\langle H \rangle\rangle$, i.e. the Tannakian data of Paper VI.*

Remark 4.8 (Averaging over lifts does not help). One may make the model canonical by adjoining the lift to the unit space: replace G by $G \times \prod_i G$ with Haar measure, let the i -th extra coordinate k_i be a conjugator, and use the lifts $k_i u_i k_i^{-1}$ in (4.2). The extra coordinates are not moved by any arrow, so the ergodic decomposition is fibred over the space of lifts, the KMS simplex is the space of probability measures on the bundle of coset spaces $H_k \backslash G$ over $k \in \prod G$, and the symmetry G acts within the fibres: the extreme points are not a single G -orbit. Canonicity is bought at the price of the transitivity that defines symmetry breaking.

4.6. When the lift does not matter.

Proposition 4.9 (Rigid Tannakian subgroups). *If N_β has no proper closed subgroup whose normal closure in G is N_β — for instance if N_β is trivial, or of prime order, or more generally if every proper closed subgroup of N_β is contained in a proper closed G -normal subgroup of N_β — then $H = N_\beta$ for every lift, and $\text{KMS}_\beta(\mathcal{A}_c) \cong \text{Prob}(G/N_\beta)$ with the transitive G -action, for every lift.*

Proof. By Corollary 4.4, $H \leq N_\beta$ and $\langle\langle H \rangle\rangle = N_\beta$; the hypothesis leaves only $H = N_\beta$. $\square$

Example 4.10 (A nontrivial $\text{Prob}(G/N_\beta)$ realized for every lift). Let $G = S_3$ and split $S = S_1 \sqcup S_2$ with $\sum_{p \in S_1} Np^{-\beta} = \infty$ and $\sum_{p \in S_2} Np^{-\beta} < \infty$; assign the class of 3-cycles to the primes of S_1 and the class of transpositions to those of S_2 . Then $\Xi_\beta(\text{sgn}, S) = 4 \sum_{S_2} Np^{-\beta} < \infty$, since the sign is trivial on 3-cycles, while $\Xi_\beta(\rho, S) \geq \sum_{S_1} Np^{-\beta} \|I - \rho((123))\|_{\text{HS}}^2 = 6 \sum_{S_1} Np^{-\beta} = \infty$ for the two-dimensional ρ . So $\mathcal{T}_\beta = \{\mathbf{1}, \text{sgn}\} = \text{Rep}(S_3/A_3)$ and $N_\beta = A_3$, of prime order; by Proposition 4.9, for every lift the ordered model has exactly two extremal KMS_β states, exchanged by the transpositions of S_3 and fixed by A_3 : the simplex is $\text{Prob}(S_3/A_3)$, a non-abelian symmetry group breaking to a normal subgroup, as the Tannakian formalism predicts. Chebotarev's theorem, used as in the constructions of [7, §5], supplies such splittings from an S_3 -extension of K : the primes whose Frobenius class consists of 3-cycles have density $\frac{1}{3}$, so their Dirichlet sum diverges for $\beta \leq 1$, and for S_2 any set of transposition primes with convergent sum will do.

5. VERDICT

Route or statement	Outcome	Where
B: $\mathcal{B} \rtimes J_S$, $\mathcal{B}$ noncommutative	no-go: KMS states are traces on $\mathcal{B}$	2.1
A: isotropy = N_β	no-go: gives $\text{Irr}(N_\beta)$ and a fixed point	3.1
A: skew product by c	$\text{Prob}(H \backslash G)$, H the Mackey range	4.2
$\rho \circ c$ coboundary $\iff \Xi_\beta(\rho, S) < \infty$	true, every lift	4.3
$N_\beta = \langle\langle H \rangle\rangle$; $\text{Prob}(G/N_\beta) \iff H \trianglelefteq G$	true	4.4
mixing means $\Rightarrow H = G$	true	4.5
simplex independent of the lift	false (S_3)	4.6
$\text{Prob}(G/N_\beta)$ for every lift	N_β rigid, e.g. A_3	4.9, 4.10

The decisive statements are these. There is no non-abelian Bost–Connes system in which the representation theory of the coefficient algebra or the isotropy of the groupoid produces the symmetry breaking: Theorems 2.1 and 3.1 are no-go theorems in the strict sense, for every $\mathcal{B}$ and every $\mathcal{G}$. There is a non-abelian Bost–Connes groupoid, principal, étale and amenable, in which the

Frobenius classes drive the dynamics through a cocycle with non-commuting increments, and its equilibrium theory is completely determined: the simplex is $\text{Prob}(H \backslash G)$, the Frobenius Dirichlet series decides which representations are coboundaries by the non-abelian Kakutani theorem, and N_β is the normal closure of H . What the Frobenius classes do not determine is H itself. Symmetry breaking to G/N_β is realized by this groupoid exactly when the lifts mix enough to make H normal, always when N_β is rigid, and, when the lifts are chosen to commute as in Paper VII, exactly when the abelian essential-value group E_β happens to be normal. A canonical non-abelian system, one whose equilibrium states depend on the Frobenius classes alone, does not exist within this framework; the canonical object is N_β , and the passage from N_β to a simplex requires a choice that the arithmetic does not make.

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